

## SAFETY DATA SHEET

Creation Date 18-September-2014

Revision Date 24-December-2021

Revision Number 5

### 1. Identification

**Product Name** DL-1,4-Dithiothreitol

**Cat No. :** AC165680000

**CAS-No** 3483-12-3  
**Synonyms** Cleland's reagent; DTT; threo-1,4-Dimercapto-2,3-butanediol

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Acros Organics BVBA  
 Janssen Pharmaceuticaaan 3a  
 2440 Geel, Belgium

##### **Manufacturer**

Fisher Scientific Company  
 One Reagent Lane  
 Fair Lawn, NJ 07410  
 Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
 Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |            |
|---------------------------------------------------------|------------|
| <b>Acute oral toxicity</b>                              | Category 4 |
| <b>Skin Corrosion/Irritation</b>                        | Category 2 |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 2 |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3 |
| Target Organs - Respiratory system.                     |            |
| <b>Combustible Dusts</b>                                | Category 1 |

#### Label Elements

**Signal Word**  
 Warning

**Hazard Statements**  
 May form combustible dust concentrations in air

Harmful if swallowed  
Causes skin irritation  
Causes serious eye irritation  
May cause respiratory irritation



### Precautionary Statements

#### Prevention

Keep container tightly closed  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

#### Response

In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion  
IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Call a POISON CENTER/ doctor if you feel unwell  
Rinse mouth  
Take off contaminated clothing

#### Storage

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

#### Disposal

Dispose of contents/container to an approved waste disposal plant

### Other Hazards

Stench

## 3. Composition/Information on Ingredients

| Component                                 | CAS-No    | Weight % |
|-------------------------------------------|-----------|----------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 3483-12-3 | >95      |

## 4. First-aid measures

|                     |                                                                                                                                                                                                                                                                                                              |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>  | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                              |
| <b>Skin Contact</b> | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention if symptoms occur.                                                                                                                                                                                                  |
| <b>Inhalation</b>   | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical attention. If not breathing, give artificial respiration. |
| <b>Ingestion</b>    | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                               |

**Most important symptoms/effects** No information available.  
**Notes to Physician** Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media** Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** > 110 °C / > 230 °F

**Method -** No information available

**Autoignition Temperature** Not applicable

### Explosion Limits

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

### Specific Hazards Arising from the Chemical

Fine dust dispersed in air may ignite. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Sulfides.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
1

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation. Avoid contact with skin, eyes or clothing.

**Environmental Precautions** Avoid release to the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Avoid contact with skin, eyes or clothing. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.** Keep in a dry place. Keep container tightly closed. Keep under nitrogen. Keep refrigerated. Incompatible Materials. Bases. Metals. Reducing Agent.

## 8. Exposure controls / personal protection

**Exposure Guidelines** This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies.

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection**  
**Hand Protection**

Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### **Respiratory Protection**

No protective equipment is needed under normal use conditions.

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                               |                                      |
|-----------------------------------------------|--------------------------------------|
| <b>Physical State</b>                         | Powder Solid                         |
| <b>Appearance</b>                             | White                                |
| <b>Odor</b>                                   | Stench                               |
| <b>Odor Threshold</b>                         | No information available             |
| <b>pH</b>                                     | 5.1 (1 %)                            |
| <b>Melting Point/Range</b>                    | 39 - 44 °C / 102.2 - 111.2 °F        |
| <b>Boiling Point/Range</b>                    | 125 - 130 °C / 257 - 266 °F @ 2 mmHg |
| <b>Flash Point</b>                            | > 110 °C / > 230 °F                  |
| <b>Evaporation Rate</b>                       | Not applicable                       |
| <b>Flammability (solid,gas)</b>               | No information available             |
| <b>Flammability or explosive limits</b>       |                                      |
| Upper                                         | No data available                    |
| Lower                                         | No data available                    |
| <b>Vapor Pressure</b>                         | No information available             |
| <b>Vapor Density</b>                          | Not applicable                       |
| <b>Specific Gravity</b>                       | No information available             |
| <b>Solubility</b>                             | completely soluble                   |
| <b>Partition coefficient; n-octanol/water</b> | No data available                    |
| <b>Autoignition Temperature</b>               | Not applicable                       |
| <b>Decomposition Temperature</b>              | No information available             |
| <b>Viscosity</b>                              | Not applicable                       |
| <b>Molecular Formula</b>                      | C4H10O2S2                            |

Molecular Weight

154.25

## 10. Stability and reactivity

|                                         |                                                                                                  |
|-----------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                                       |
| <b>Stability</b>                        | Moisture sensitive. Air sensitive.                                                               |
| <b>Conditions to Avoid</b>              | Avoid dust formation. Incompatible products. Excess heat. Exposure to air. Exposure to moisture. |
| <b>Incompatible Materials</b>           | Bases, Metals, Reducing Agent                                                                    |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Sulfur oxides, Sulfides                 |
| <b>Hazardous Polymerization</b>         | No information available.                                                                        |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                    |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component                                 | LD50 Oral         | LD50 Dermal | LC50 Inhalation |
|-------------------------------------------|-------------------|-------------|-----------------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 400 mg/kg ( Rat ) | Not listed  | Not listed      |

**Toxicologically Synergistic Products** No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | Irritating to eyes, respiratory system and skin                                          |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                 | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------------------------------|-----------|------------|------------|------------|------------|------------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 3483-12-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                                                   |                                                                |
|---------------------------------------------------|----------------------------------------------------------------|
| <b>Mutagenic Effects</b>                          | No information available                                       |
| <b>Reproductive Effects</b>                       | No information available.                                      |
| <b>Developmental Effects</b>                      | No information available.                                      |
| <b>Teratogenicity</b>                             | No information available.                                      |
| <b>STOT - single exposure</b>                     | Respiratory system                                             |
| <b>STOT - repeated exposure</b>                   | None known                                                     |
| <b>Aspiration hazard</b>                          | No information available                                       |
| <b>Symptoms / effects, both acute and delayed</b> | No information available                                       |
| <b>Endocrine Disruptor Information</b>            | No information available                                       |
| <b>Other Adverse Effects</b>                      | The toxicological properties have not been fully investigated. |

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

**UN-No** UN3335  
**Proper Shipping Name** AVIATION REGULATED SOLID, N.O.S.  
**Hazard Class** 9

### TDG

**UN-No** UN3335  
**Proper Shipping Name** AVIATION REGULATED SOLID, N.O.S.  
**Hazard Class** 9

### IATA

**UN-No** UN3335  
**Proper Shipping Name** AVIATION REGULATED SOLID, N.O.S.\*  
**Hazard Class** 9

### IMDG/IMO

Not regulated

## 15. Regulatory information

### International Inventories

| Component                                 | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-------------------------------------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 3483-12-3 | X   | -    | X    | ACTIVE                                        | 222-468-7 | -      | -   |

| Component                                 | CAS-No    | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-------------------------------------------|-----------|-------|------|------|------|------|------|-------|-------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 3483-12-3 | X     | -    | -    | -    | X    | X    | X     | X     |

### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous

Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                 | CAS-No    | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------------------|-----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 3483-12-3 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                 | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2,3-Butanediol, 1,4-dimercapto-, (R*,R*)- | 3483-12-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

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**Creation Date**

18-September-2014

**Revision Date**

24-December-2021

**Print Date**

24-December-2021

**Revision Summary**

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**